MIRAI AI INTERVIEWER

MSOT Pilot Feasibility & Economics | Judge-ready decision brief | 20 October 2026

Readiness verdict: Tested offline workflow; NOT ready for a 350-student production voice rollout. Real voice, recording, institutional login, scoring validation and 50/100/150 audio load tests are incomplete.

The 350-student challenge in four numbers

Cohort	Total interview time	At 150 concurrent	Illustrative planning budget
200 students	50 hours	2 x 15-minute slots	~INR 10,600
250 students	62.5 hours	2 x 15-minute slots	~INR 13,250
300 students	75 hours	2 x 15-minute slots	~INR 15,900
350 students	87.5 hours	3 x 15-minute slots	~INR 18,550

Slot times are arithmetic only, excluding admission/changeover/recovery. Costs are modeled, not measured; fixed plans, taxes, staffing and additional infrastructure are excluded.

Illustrative cost per 15-minute viva: ~INR 53

Line item	Assumption	INR per viva
Deepgram Nova-3 STT	15 min at regular $0.0092/min; FX 90	12.42
Sarvam Bulbul v3 TTS	5,000 chars at INR 30 / 10,000	15.00
Gemini 3 Flash via LiveKit	30k input + 3k output tokens	2.16
LiveKit hosted agent (illustrative)	15 min at $0.01/min; replace if self-hosted	13.50
Recording, storage and egress	Unverified allowance	3.00
Contingency	15% on INR 46.08 subtotal	6.91
TOTAL	Planning scenario; NOT a provider quote	52.99

Sensitivity: lean ~INR 34/viva; planning ~INR 53; stress ~INR 75. Uses an assumed USD/INR of 90, not a live FX quote. Provider pricing and hosting topology must be confirmed before use.

Source rates: deepgram.com/pricing | docs.sarvam.ai/api/getting-started/pricing | livekit.com/pricing/inference | livekit.com/pricing

CAPACITY, ARCHITECTURE & PROOF

What would be required for an MSOT deployment | Proposed only, not implemented

Cohort schedule and external quotas

Concurrent capacity	350-student schedule	Important limitation
50 sessions	7 slots = 105 min	Fails under-1-hour target
100 sessions	4 slots = 60 min	No time for changeover
150 sessions	3 slots = 45 min	Under 1 hour only if overhead < 15 min

External blocker: LiveKit Inference public Build/Ship/Scale concurrency allowances are 5/20/50. A 150-session cohort needs custom quota approval or a different validated arrangement. Deepgram lists up to 150 WebSocket streams on Pay As You Go, leaving no headroom; Sarvam account-specific rate limits remain unverified.

Two separate architectures — do not confuse them

VERIFIED OFFLINE PROTOTYPE	PROPOSED LIVE/PRODUCTION SYSTEM
Faculty fixture -> approved bank -> student consent -> deterministic interview -> transcript digest -> synthetic scoring -> reasoned faculty override	Student WebRTC -> admission/consent -> isolated agent worker -> streaming STT/LLM/TTS -> incremental two-party recording + durable state -> queued scoring -> faculty replay/override

The current voice PoC is gated to one approved session, <=120 seconds and five turns. The proposed right-hand pipeline, storage, queue, and high concurrency are not live-verified.

Evidence required before a 350-student pilot

Gate	Minimum proof	Status
Real voice + audit	5 recorded 15-minute vivas, transcripts, scores and faculty audio replay	MISSING
Scoring and fairness	>=10 human/AI score pairs, repeated scoring, prompt injection/fairness tests	MISSING
Speech quality	WER on >=20 Indian English/Hinglish clips; p50/p95; p95 <2.5s	MISSING
Scale and recovery	50/100/150 real audio bots; errors, drops, CPU, breaking point	MISSING
Institutional readiness	Quotas, budget cap, auth, encrypted durable storage, retention policy	MISSING

What is genuinely demonstrated: 64 frontend/browser tests, 28 Python tests, TypeScript/build checks and offline faculty-to-student walkthrough (agent-reported with published logs). These are synthetic/mocked tests, not real voice or scale results.

Full evidence and assumptions: docs/submission/TEST_REPORT.md, COST_SHEET.md, CAPACITY_PLAN.md, ARCHITECTURE.md and PRIVACY_NOTE.md.